# Guillermo Fernando Farfán Romero

## Senior Full Stack Developer (React / TypeScript / Python / Node.js)

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## PROFESSIONAL SUMMARY

Senior Full Stack Developer and Technical Lead with 8+ years of experience engineering reactive web applications, real-time analytical dashboards, and high-performance microservices architectures. Specialist in modern frontend engineering with **React.js (React 18/19), Next.js, Vite, TypeScript (Strict Mode), TailwindCSS, and TanStack Query**, paired with robust backends in **Python (FastAPI, Django REST Framework, SQLAlchemy 2.0, Pydantic v2)**, **Node.js (Express, Prisma ORM)**, and **Java (Spring Boot)**, backed by **PostgreSQL (with pgvector), SQL Server, and Redis**.

Proven track record optimizing **Core Web Vitals (Lighthouse >90)**, integrating **GenAI / LLMs (Google Gemini API, Vertex AI, OpenAI API, RAG)**, enforcing web accessibility (**WCAG 2.1**), automating **Zero-Downtime deployments on Google Cloud (Cloud Run, GKE)**, and mentoring developers as a former Frontend Instructor at **Coderhouse**.

## TECHNICAL SKILLS & COMPETENCIES

- **Frontend Core & UI:** React.js (React 18/19), Next.js (App Router, SSR/SSG), Vite, TypeScript (Strict Mode), JavaScript (ES6+), TailwindCSS, TanStack Query (React Query), Redux Toolkit, Zustand, Semantic HTML5, CSS3, Design Systems, Web Accessibility (WCAG 2.1 / a11y), Core Web Vitals & Lighthouse Optimization (>90).
- **Backend, APIs & GenAI:** Python 3.10+ (FastAPI, Django REST Framework - DRF, Pydantic v2, SQLAlchemy 2.0, Asyncio, Celery), Node.js (Express, Prisma ORM, WebSockets / SSE), Java (Spring Boot), GenAI / LLM Integration (Google Gemini API, Vertex AI, OpenAI API, RAG, pgvector), RESTful APIs, OpenAPI/Swagger, JWT / OAuth2, Zod.
- **Databases & Caching:** PostgreSQL (Relational modeling, pgvector, Prisma / Alembic), Redis (Distributed Caching, Pub/Sub, Rate Limiting), Microsoft SQL Server, MySQL, MongoDB, Baserow.
- **Cloud, DevOps & Automation:** Google Cloud Platform (Cloud Run, GKE Autopilot, Cloud Storage, Firebase/CDN), Docker, Docker Compose, Terraform, GitHub Actions, GitLab CI, ArgoCD, n8n.
- **Testing, Quality & Methodologies:** Jest, React Testing Library, Playwright, Pytest, SonarQube, Clean Architecture, Domain-Driven Design (DDD), Scrum, Code Reviews, Architecture Decision Records (ADRs).

## PROFESSIONAL EXPERIENCE

**Senior Python Backend & Financial Automation Engineer** | Reforest Latam

*May 2026 – Present | Full-time (Remote)*

- Architected and developed end-to-end the financial operations platform and transactional Telegram bot in **Python 3.11+ (aiogram 3.x, asyncio, Pydantic)** with **Redis-persisted FSM** state management (RedisStorage and events_isolation double-click protection), shared HTTP connection pooling (aiohttp), exact decimal accounting (ROUND_HALF_UP), real-time **fail-closed RBAC**, and **Baserow / Google Drive API** integration.
- Built an **atomic multi-currency bank reconciliation engine** featuring native PDF/CSV/Excel statement parsers (BBVA, Galicia, Macro, Mercury, Balanz), **SHA-256** deduplication in Redis, decoupled staging with semantic auto-classification, batch injection with **atomic batch rollback**, historical **MEP Dollar** exchange rate lookup with T-1 to T-5 business-day fallback, and sanitized **ARCA** tax exports.
- Implemented **dual LLM OCR & automated classification (**Gemini 2.5 Flash **and** DeepSeek **via OpenRouter)** using semaphore-bounded concurrent PDF chunking and prompt-injection sanitization; deployed on **Linux VPS with Docker Compose**, automated daily Drive backups, and **GitHub Actions CI/CD** running **Ruff** and **514 passing Pytest integration tests (100% green)**.

**Senior Full Stack & Software Consultant** | Independent Consulting & Enterprise Projects

*January 2021 – Present | Strategic B2B Technology Consultant (Independent / Contract / Remote)*

- **Fintech, Trading & E-Commerce Platforms (Clients under NDA):** Designed and deployed web applications, high-concurrency REST APIs, and credit-scoring microservices using **React, Next.js, TypeScript, FastAPI, Django REST Framework, and Node.js**, enforcing strict schema validation with **Pydantic/Zod**, OpenAPI/Swagger contracts, **JWT/OAuth2** authentication, and query optimization with **SQLAlchemy 2.0 and Redis (>60% p95 latency reduction and Lighthouse >90)**.
- **Serverless AI Financial Analytics Platform & Industrial Telemetry:** Architected a 100% serverless full-stack platform with **React 19 + Vite (CDN), Node.js, PostgreSQL with Prisma ORM, and native Google Gemini API integration** for automated insight generation, alongside operational dashboards and **Java/Python services on GCP**.
- **Enterprise Portals, Cloud & Observability (Clients under NDA):** Led the cloud modernization and deployment to **GCP (GKE, Cloud Run, and Terraform)** of transactional portals and containerized Docker services, and instrumented end-to-end observability (from frontend to SQL query) with **OpenTelemetry, Prometheus, and Grafana (-45% MTTR)**.

**Senior Software Engineer & Database Administrator (DBA)** | University of Buenos Aires (UBA)

*March 2018 – February 2024 | Institutional Systems Specialist (Part-time / Buenos Aires, Argentina)*

- Developed responsive web interfaces and analytical dashboards (**React, TypeScript, JavaScript ES6+**) integrated with **Python** APIs and ETL pipelines, **slashing complex report generation times from minutes to seconds** for a community of **+50,000 users (+10,000 concurrent active users)**.
- Optimized the persistence layer across **8+ TB of institutional data (+500 GB migrated)** on **PostgreSQL and SQL Server**, achieving a **55% reduction in p95 query latency (-40% average response time)** with **99.9% uptime**.
- *Concurrent Frontend Teaching:* **Technical Tutor / Frontend Instructor (React.js & JavaScript)** at **Coderhouse** *(Jul 2022 – Sep 2022)*, mentoring **+50 students** in React.js, JS ES6+, HTML5, CSS3, Git, and REST API integration (**95% course completion/approval rate**).

**Systems Infrastructure & Operations Coordinator** | Government of the City of Buenos Aires (GCBA)

*January 2020 – December 2023 | Ministry of Economic Development (Full-time / Buenos Aires, Argentina)*

- Standardized accessible digital form architecture (**WCAG 2.1**) and automated backend validation layers, achieving a **30% reduction in feature delivery time**, **-40% data inconsistencies**, and **-25% to -30% production incidents** across citizen portals serving **100,000+ monthly users (99.9% SLA)** on **SQL Server, PostgreSQL, MySQL, and MongoDB**.

## FEATURED PROJECTS & HACKATHONS

- **Monitor CI/CD — Real-Time DORA Metrics Dashboard:** Full-stack platform built with **TypeScript, Node.js, React, and WebSockets** providing live pipeline telemetry for GitHub Actions and GitLab CI under DORA standards.
- **Autonomous FinOps Engine (Aleph Hackathon 2026) & OmniSync (EDA):** Autonomous Python + Terraform engine for GCP cost optimization, and an event-driven microservices engine featuring Dead Letter Queues (DLQ).

## EDUCATION, CERTIFICATIONS & LANGUAGES

- **Education: Associate Degree in Information Systems** | UTN *(In Progress — Expected Graduation: Dec 2027)* · **Bachelor's Studies in Labor Relations & HR** | UBA *(Advanced University Coursework Completed / Organizational Leadership)*.
  - *Certifications* (Verifiable on Credly & Coursera):* Meta Full-Stack Engineer Certificate | IBM Full Stack Software Developer Professional Certificate (V5) | Meta Front-End Developer Certificate | Meta Back-End Developer Certificate | IBM JavaScript Backend Developer (V2) | IBM DevOps and Software Engineering.
- **Languages: Spanish:** Native · **English:** Professional Working Proficiency (B2 — Fluent technical communication for live engineering syncs, RFCs, and documentation).